

STAC67: Regression Analysis

Lecture 3

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Simple linear model with normal errors

- The random errors are sometimes assumed to be normally distributed.
- Simple linear model with **normal errors**:

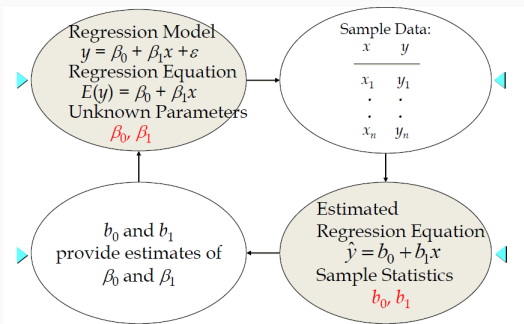
$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i,$$

where ϵ_i are independent and identically distributed (i.i.d) with normal distribution with mean 0 and variance σ^2 .

- In terms of Y, this means that we have the conditional distribution

$$Y|X = x \sim N(\beta_0 + \beta_1 x, \sigma^2)$$

Parameter Estimation



- Maximum likelihood estimation

$$f(y; \beta_0, \beta_1, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \beta_0 - \beta_1 x_i)^2\right)$$

Likelihood Function:

$$L(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n f(y_i; \beta_0, \beta_1, \sigma^2)$$

Parameter Estimation

To find m.l.e for β_0, β_1 , and σ^2 , we can maximize the following:

Log-likelihood Function:

$$\ln L(\beta_0, \beta_1, \sigma^2) = K - n \ln \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

Maximizing Likelihood function w.r.t β_0, β_1 is equivalent to minimizing,

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

Parameter Estimation

Method of least squares

Simple linear model:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Goal: Find the **best** estimates, $\hat{\beta}_0$ and $\hat{\beta}_1$ given the data.

- What does it mean, “**best**”?
- Least squares: best by criterion

$$Q(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

- Find the $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimizes the criterion Q .
 1. Write the normal equations (derivatives of Q set to 0)
 2. Find the solution of the normal equations

Least Square Estimators

Other criteria. Why square the errors?

We could use least absolute deviations estimates, minimizing

$$Q_1(\beta_0, \beta_1) = \sum_{i=1}^n |(y_i - \beta_0 - \beta_1 x_i)|$$

- **Convenience**
- **Optimality**

Theorem 1 (Gauss-Markov)

Consider the simple linear model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Suppose that the following assumptions (called **Gauss-Markov assumptions**) concerning the random errors are satisfied:

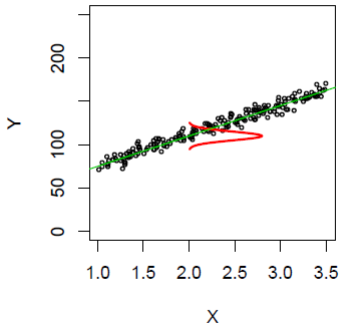
- Mean zero: $E(\epsilon_i) = 0$
- Constant variance: $Var(\epsilon_i) = \sigma^2$
- Uncorrelated: $Cov(\epsilon_i, \epsilon_j) = 0, \quad i \neq j$

Then the least square estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased and have minimum variance among all unbiased linear estimators (**BLUE**).

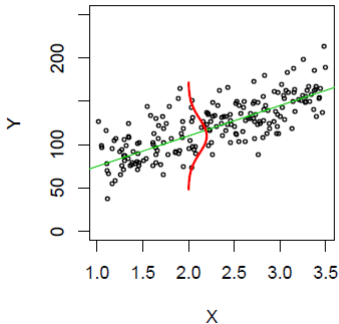
The interpretation of σ^2

The variance, σ^2 controls the **dispersion** of Y around $\beta_0 + \beta_1 X$

small dispersion



large dispersion



Fitted values and Residuals

Definition

Regression equation or fitted regression line

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X,$$

where $\hat{Y}$ is the estimated mean of the response variable at level X of the explanatory.

- For each observation, we can compute the **fitted value**:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i, \quad i = 1, \dots, n$$

- The vertical distance from the observed y_i to the fitted value is called the **residual**

$$e_i = Y_i - \hat{Y}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i), \quad i = 1, \dots, n$$

The residuals can be thought of as predicted values of the unknown errors $\epsilon_1, \dots, \epsilon_n$.